M15 | Water-hydraulic motor

Compact, water-lubricated axial-piston technology for a defined operating duty.

M15 converts water pressure into shaft torque and rotary motion. Size the starting load separately from the continuous duty, and match flow to speed and displacement.

Published parameter	M15 range rating
Maximum displacement (cm³/rev)	18.6
Minimum displacement (cm³/rev)	15.0
Maximum speed (rpm)	4000
Minimum speed (rpm)	500
Maximum continuous power (kW)	17.5
Maximum input flow (L/min)	72
Maximum continuous pressure (bar)	160
Mass (kg)	6
Maximum temperature (°C)	50
Minimum temperature (°C)	2

Nominal ordering displacements: 12 / 15 / 19 cm³/rev. Nominal options and range-table limits are different source entries; use the selected configuration.

Design & environmental advantages

Multiple pistons smooth the delivery of power. Water lubricates the working surfaces, removing hydraulic oil from this circuit. Standard stainless construction and selectable internal materials support fluid-specific selection.

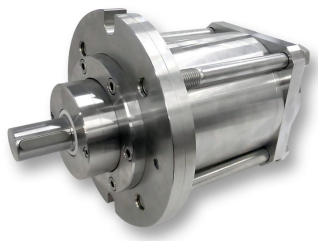
Efficiency & selection

Match component size to useful work. Read the supplied efficiency curve at pressure and speed. Avoid sustained relief flow and excessive throttling; these convert useful hydraulic power into heat.

Starting, overrun and procurement

Manufacturer starting-load guidance is 60% of dynamic torque. M3/M6 need at least 25 bar unloaded; larger models are stated to start above 12 bar. Specify load inertia, direction, case drain and overrun protection. Support radial and axial loads externally. Variable versions: M60 and M180.

Maxima are separate catalogue limits. Obtain the exact configuration drawing, duty-point selection, seal/service schedule, material documentation and applicable approval evidence with the quotation.



Family photograph; configuration varies.

Connect the selected design

Two-port motors are unidirectional; the return line maximum is 2.5 bar. Three-port models provide a separate case drain; return pressure must exceed leakage-port pressure, which is limited to 2.5 bar.

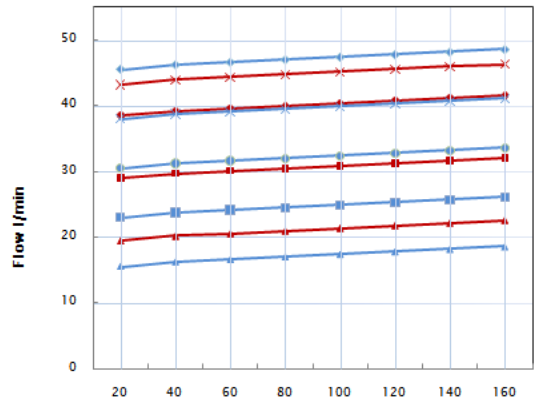
Reference conversions: 160 bar = 2,321 psi; 1 L/min = 0.264172 US gpm; 1 kg = 2.20462 lb.

Axial-Piston-Water-Motor-Full-Range-V.01.25.pdf, pp. 2-4. Website product context checked 17 Oct 2026.

M15 | Performance curves

Read displacement, speed and pressure together. Published curve data are preserved.

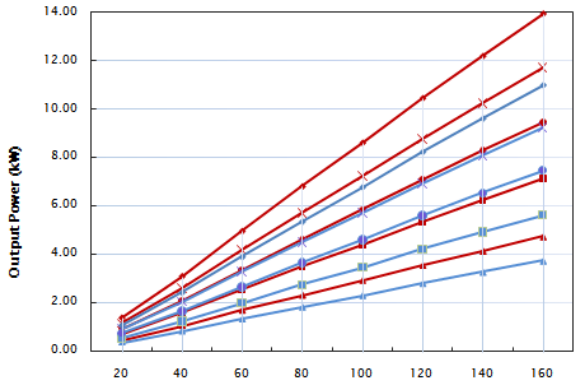
Input flow (L/min)



Pressure (bar)

19cc 1000rpm 19cc 1500rpm 19cc 2000rpm 19cc 2500rpm 19cc 3000rpm
15cc 1000rpm 15cc1500rpm 15cc 2000rpm 15cc 2500rpm 15cc 3000rpm

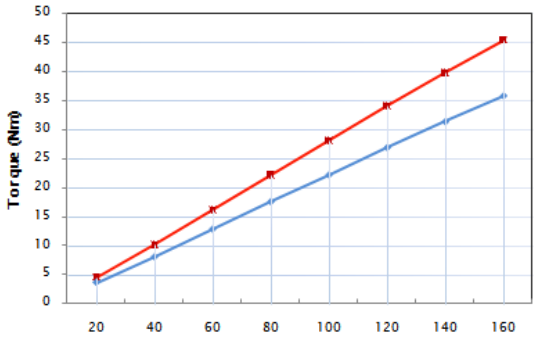
Output power (kW)



Pressure (bar)

19cc 1000rpm 19cc 1500rpm 19cc 2000rpm 19cc 2500rpm 19cc 3000rpm
15cc 1000rpm 15cc1500rpm 15cc 2000rpm 15cc 2500rpm 15cc 3000rpm

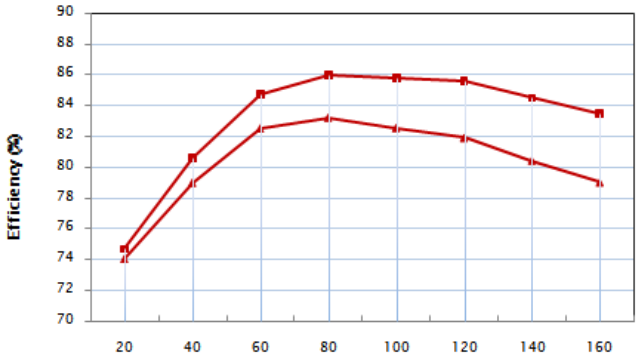
Output torque (N·m)



Pressure (bar)

19cc 1500rpm 15cc1500rpm

Overall efficiency (%)



Pressure (bar)

19cc 1000rpm 19cc 1500rpm

Apply the curves to the duty

The source input-flow legend includes 19 cm³/rev at 3,000 rpm, but that series is not clearly visible. Do not infer an additional curve.

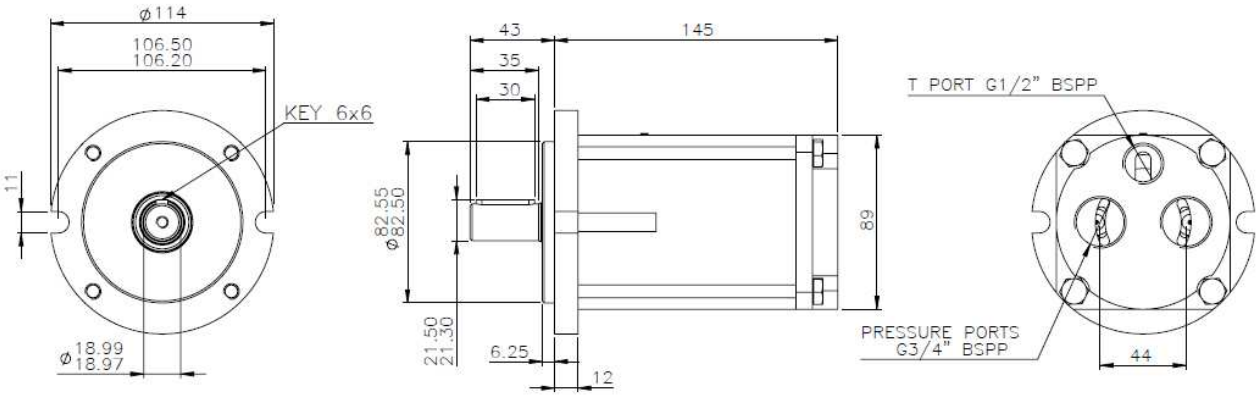
1 L/min = 0.264172 US gpm; 1 bar = 14.5038 psi; 1 kW = 1.34102 hp; 1 N·m = 0.737562 lbf·ft.

Axial-Piston-Water-Motor-Full-Range-V.01.25.pdf, p. 10.

M15 | Dimensions & connections

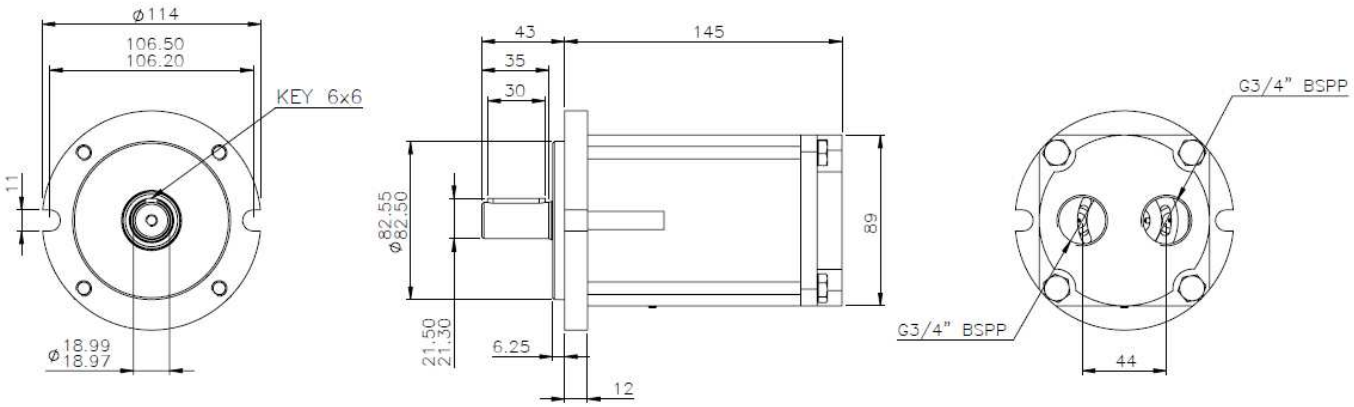
Manufacturer technical reference | dimensions govern; reproduction is not to scale.

MB160-15W & MB160-19W



MC160-15W & MA160-19W

MC160-15W & MA160-19W



Port threads and flange classes are connection designations, not converted diameters or operating-pressure ratings. Select the exact configuration shown on the approved order drawing.